

Experience

Software Engineer

HYPERSONIX INC. | PLATFORM TEAM

Bengaluru

Aug. 2020 - Present

- Designed and implemented the entire Authentication system
- Improved API performance by about 40% by optimizing the codebase and incorporating compression algorithms like Brotli
- Set up logging mechanism for platform codebase using ELK and Filebeat
- Developed multiple critical customer-facing components like Hints, Guided Tours, and Learning Center

Researcher

GEORGIA INSTITUTE OF TECHNOLOGY | MASTER'S PROJECT & SPECIAL PROBLEMS

Atlanta, USA

Jan. 2020 - Present

- Working under **Dr. Thad Starner** to develop tools to Counter Plagiarism and assist online Education at Scale.
- Our research project **Jack Watson** got featured in **Forbes**, **TimesHigherEducation**, and multiple other places.
- Ongoing research projects include -
 - **Jack Watson**: An AI agent that monitors Freelancer websites and tries to catch students if they hire someone to do their work.
 - **Examinator**: A similarity detection tool to catch students who collaborate or cheat in take-home exams.
 - **JackMarker**: A tool to watermark code/text and detect plagiarism at scale.
 - **GitDown**: A tool to monitor public Github repositories and submit take down requests to students.

Graduate Programmer

FIDELITY INTERNATIONAL | BEHAVIORAL FINANCE TEAM

Gurugram

June. 2018 - Dec 2018

- Developed a real-time feedback application using web sockets, Django, Redis, and D3.js.
- Worked in the behavioral finance team and helped in designing tools for portfolio managers using React.

Skills

Languages: Python, Javascript, C++(basic); **Web:** Django, Flask, React, PostgreSQL, Redis, ELK, Auth0, JSON-LD, Unit testing, Celery, Web Sockets, gRPC, Semantic Web ; **AI:** Numpy, Pandas, Sklearn, OpenCV, Tensorflow, Pytorch, Fast.ai; **Other:** Algorithms, Data Structures, Docker, AWS, GCP, Linux, Inkscape, UX Research, UI Design, Web scraping

Education

Master of Science | Computer Science

GEORGIA INSTITUTE OF TECHNOLOGY | GPA 4.0/4.0

Atlanta, USA

Jan 2019 - Exp. May 2021

- Machine Learning Specialization with Research in Counter Plagiarism Technology

Bachelor of Technology | Electronics and Communication Engg.

DCRUST | GPA 7.05/10

Murthal, India

July 2014 - Apr. 2018

- Participated in numerous Robotics Competitions across multiple IITs

Relevant Coursework

Georgia Tech: Artificial Intelligence, Machine Learning, Computer Vision, Reinforcement Learning, AI Ethics and Society, Data and Visual Analytics, Knowledge-Based AI, Information Security and Policies, Graduate Algorithms, Intro to Operating Systems

Udacity Nanodegrees: Computer Vision, React Developer, FullStack Web Developer, React Developer

Fast.ai: Practical Data Ethics, Practical Deep Learning for Coders

Volunteer and Other Experience

2018-21 **Organization Admin and Mentor**, HydraEcosystem.org

Google Summer of Code

2017-20 **Mentor and Project Reviewer**, Udacity

Remote

2018 **Tech Speaker**, PyCon India

Hydrabad, India

2017 **Participant**, Python Software Foundation

Google Summer of Code

2015-17 **Founder**, Typinggeek.com

Delhi

Publications

1. Raghav Apoorv, Akshay Dahiya, Uma Sreeram, Bharat Rahuldhev Patil, India Irish, Rocko Graziano, and Thad Starner. 2020. Examinator: A Plagiarism Detection Tool for Take-Home Exams. In Proceedings of the Seventh ACM Conference on Learning @ Scale(L@S '20). Association for Computing Machinery, New York, NY, USA, 261–264. DOI: <http://dx.doi.org/10.1145/3386527.3406723>